



Multi-Method Convergence Protocol for Robust Multi-Criteria Ranking in Controlling Strategies of Photovoltaic Systems

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Abstract: This paper proposes a Multi-Method Convergence Protocol (MMCP) for the robust ranking of 16 Maximum Power Point Tracking (MPPT) strategies applied to a photovoltaic (PV)–boost system. Unlike single-criterion comparisons, the approach integrates a hybrid objective weighting scheme based on Criteria Importance Through Intercriteria Correlation (CRITIC)–entropy, four complementary Multi-Criteria Decision-Making (MCDM) methods—Technique for Order Preference by Similarity to Ideal Solution (TOPSIS), Preference Ranking Organization Method for Enrichment Evaluation II (PROMETHEE II), VlseKriterijumska Optimizacija I Kompromisno Resenje (VIKOR), and Elimination and Choice Translating Reality II (ELECTRE II), and a consensus-based aggregation using Borda–Copeland rules. The proposed MMCP results reveal a stable and highly convergent ranking. The final aggregation ranks the Fuzzy Logic controller first, with a Borda score of 60, a Copeland score of 15, and a total score of 75, followed by the standalone artificial neural network (ANN)-based MPPT and the sliding mode control–artificial neural network (SMC–ANN) method. Intermediate positions are occupied by sliding mode control 1st order (SMC1), sliding mode control 2nd order (SMC2), and Incremental Conductance (INC)-based MPPT optimized by Whale Optimization Algorithm (INC–WOA), while Perturb and Observe (P&O)-based MPPT Algorithm and INC-based MPPT Algorithm rank last. The robustness of the decision-making process is confirmed by strong agreement among methods, with Spearman coefficients ranging from 0.891 to 0.997 and Kendall's coefficient of concordance of 0.9449. These findings demonstrate that the proposed MMCP framework provides a consistent, traceable, and methodologically robust global ranking, facilitating the selection of MPPT strategies based on an explicit multi-criteria trade-off of the photovoltaic system.

Keywords: Hybrid CRITIC–entropy weighting; Photovoltaic systems; Maximum Power Point Tracking; Multi-Criteria Decision-Making; Multi-Method Convergence Protocol; Rank aggregation; Robust ranking

1 Introduction

The sustained growth of photovoltaic (PV) energy in the global energy mix reinforces the need for advanced control strategies capable of maximizing energy extraction under highly variable environmental conditions. Maximum Power Point Tracking (MPPT) remains a central challenge in this regard, as the optimal operating point of a PV generator continuously shifts with irradiance and temperature, requiring the controller to dynamically adjust the duty cycle of the Direct Current to Direct Current (DC–DC) converter. Conventional methods such as Perturb and Observe (P&O) and Incremental Conductance (INC) remain widely used for their simplicity, but suffer from steady-state oscillations, slow response under rapidly changing irradiance, and difficulty locating the global maximum under partial shading.

To address these limitations, literature has proposed a wide range of hybrid and intelligent MPPT strategies. Metaheuristic-assisted variants of P&O, combining it with algorithms such as the Whale Optimization Algorithm, Grey Wolf Optimizer, Shuffled Frog Leaping Algorithm, or Bat Algorithm, improve global maximum tracking under partial shading at the expense of higher computational complexity. Artificial neural network (ANN) and fuzzy-logic-based controllers further improve convergence speed and reduce oscillations, though their performance depends on

training data quality or rule-based design. A smaller number of comparative studies have benchmarked some of these algorithms together, generally confirming that hybrid and intelligent methods outperform conventional P&O/INC, but typically using only one or two performance criteria and a simple weighted-sum ranking. This reveals a clear gap: no study has yet proposed a systematic multi-criteria framework that jointly evaluates MPPT efficiency, convergence speed, settling time, and integrated error indices using multiple weighting and ranking methods combined into a consensus result. Existing comparisons remain dependent on a single metric or a single aggregation scheme, which can produce unstable or weighting-sensitive conclusions. This gap motivates the Multi-Method Convergence Protocol (MMCP) proposed in this study, which provides a multi-criteria framework for ranking MPPT strategies by combining Criteria Importance Through Intercriteria Correlation (CRITIC)–entropy weighting with Technique for Order Preference by Similarity to Ideal Solution (TOPSIS), Preference Ranking Organization Method for Enrichment Evaluation II (PROMETHEE II), Multi-Criteria Optimization and Compromise Solution (VIKOR), and Elimination and Choice Translating Reality II (ELECTRE II), consolidated through a Borda–Copeland consensus procedure.

The MPPT literature can be organized into several research axes, ranging from general reviews and taxonomies to quantitative comparisons, intelligent/soft-computing approaches, harmonized benchmarks, partial shading condition (PSC)/global maximum power point (GMPP)-focused studies, and single-methods or ensemble Multi-Criteria Decision-Making (MCDM) applications. Table 1 summarizes these axes in terms of their main objective, weighting strategy, ranking logic, robustness/sensitivity treatment, and data/context basis, together with their main limitation and their specific contribution to the proposed MMCP framework. This synthesis highlights that, while individual elements of the MMCP methodology appear scattered across these axes, no existing study integrates them into a unified, robustness-validated multi-criteria ranking framework for MPPT algorithms.

Table 1. Classification of relevant literature for MPPT evaluation and multi-criteria framework construction

Literature Axis	Representative References	Main Objective	Criteria Weighting	Ranking Logic	Robustness/Sensitivity	Contribution to MMCP
General reviews of MPPT techniques	[1–8]	Provide a broad overview of MPPT families and compare their principles, strengths, and limitations	Mostly absent, qualitative, or implicit	Narrative comparison and taxonomic structuring	Descriptive discussion of robustness without formal sensitivity analysis	Establishes the conceptual basis for structured multi-criteria decision frameworks
Classification/taxonomy reviews	[9–13]	Classify MPPT techniques according to control structure, complexity, and operating conditions	Not explicitly defined	Categorization into families and subclasses	Limited treatment of sensitivity and robustness	Defines evaluation dimensions for MMCP (efficiency, dynamics, robustness, complexity)
Classical quantitative comparative studies	[14–21]	Compare MPPT algorithms using efficiency, response time, oscillations, and error indices	No explicit weighting	Ranking implicitly derived from metrics	Physical robustness is sometimes evaluated; decision robustness is not addressed	Motivates explicit weighting and formal multi-criteria decision modeling
Intelligent/soft computing comparisons	[22–25]	Evaluate ANN-, FL-, PSO-, and GWO-based approaches under dynamic conditions	Implicit or empirical	Scenario-dependent ranking	Good physical robustness; no formal decision robustness	Supports the integration of intelligent control and contextual robustness into MMCP

Harmonized benchmarks/systematic evaluations	[26–30]	Provide structured comparisons using common scenarios and metrics	Rarely formalized	Direct benchmarking	Partial robustness assessment; limited sensitivity analysis	Enables the construction of homogeneous decision matrices
PSC/GMPP/severe-condition studies	[31–34]	Identify algorithms capable of tracking the GMPP under partial shading conditions	Implicit weighting focused on efficiency	Performance comparison under PSC	Strong physical robustness; weak decision robustness	Introduces robustness under extreme conditions into MMCP
Single-method MCDM approaches	[35–37]	Apply a single MCDM logic, or a single dominant ranking pipeline, to select the best alternative under multiple criteria	Subjective, objective, or hybrid depending on the selected method	Single-method ranking, typically TOPSIS-based	Sensitive to the choice of weights and the ranking model	Justifies the use of multiple complementary MCDM methods in MMCP
Data-driven objective weighting	[33, 38]	Derive weights from the statistical properties of data	Objective or hybrid	Pre-ranking weighting	Sensitive to data dispersion and correlations	Basis for hybrid CRITIC–entropy weighting in MMCP
Ensemble MCDM/rank aggregation	[38]	Combine multiple rankings to obtain a consensus decision	Multiple weighting/ranking schemes	Multi-method consensus ranking	Explicit evaluation of ranking stability	Core methodological contribution of MMCP
Robustness and sensitivity analysis	[32, 38]	Assess ranking stability under variations in weights, methods, and scenarios	Multi-scenario/perturbation-based	Post-ranking validation	Explicit robustness analysis through sensitivity or concordance checks	Supports the final validation stage of MMCP

Note: MMCP = Multi-Method Convergence Protocol; MPPT = Maximum Power Point Tracking; ANN = artificial neural network; FL = Fuzzy Logic; PSO = Particle Swarm Optimization; GWO = Grey Wolf Optimizer; PSC = Partial Shading Condition; GMPP = Global Maximum Power Point; MCDM = Multi-Criteria Decision-Making; PV = photovoltaic; TOPSIS = Technique for Order Preference by Similarity to Ideal Solution; CRITIC = Criteria Importance Through Intercriteria Correlation.

Literature on MPPT algorithms has evolved progressively from descriptive reviews to comparative analyses, and more recently toward structured MCDM frameworks. General and taxonomic reviews have played a key role in stabilizing the field by classifying MPPT techniques according to their control principles, implementation complexity, and application domains [1, 7, 12]. However, these contributions remain largely qualitative and do not provide explicit criteria weighting or reproducible ranking procedures.

Quantitative comparative studies have introduced relevant technical performance indicators, including tracking efficiency, response time, oscillations, and integrated error metrics [2, 4, 20]. Despite this progress, the decision-making process is often implicit, with the “best” algorithm inferred directly from raw metrics. This approach leads to rankings that are highly sensitive to the dominant criterion and therefore lack robustness.

Research on intelligent and hybrid MPPT strategies (e.g., ANN, Fuzzy Logic (FL), and metaheuristic-based approaches) as well as studies under partial shading conditions (PSC/GMPP) have significantly improved the evaluation of controller performance in complex and dynamic environments [19, 26]. Nevertheless, these works

remain scenario-dependent and focus primarily on physical robustness, while largely overlooking the robustness of the final decision or ranking.

More recent contributions, including harmonized benchmarks and systematic reviews, have introduced partial standardization of evaluation protocols, test conditions, and performance indicators [23, 24]. Although these efforts improve comparability across studies, they remain essentially descriptive and do not incorporate formal decision consensus mechanisms or stability validation.

In parallel, methodological developments in decision-making have explored the use of single-method MCDM approaches, data-driven weighting techniques, and rank aggregation strategies. Single-method approaches (e.g., TOPSIS-based frameworks) provide structured rankings but remain sensitive to the choice of weights and model assumptions [31, 32]. Data-driven weighting methods improve objectivity by exploiting statistical properties of the data, yet they do not fully capture the complexity of the decision structure when used alone. Similarly, rank aggregation and ensemble MCDM frameworks introduce consensus-based ranking mechanisms and enable explicit robustness analysis [38], but their application to MPPT algorithm selection remains limited.

Overall, while literature provides substantial contributions in terms of classification, technical comparison, intelligent control, and benchmarking, it remains incomplete for rigorously addressing the problem of selecting the optimal MPPT algorithm. The main gap lies in the absence of an integrated framework that simultaneously combines explicit criteria weighting, multiple ranking logics, inter-method aggregation, and formal robustness validation of the final ranking.

This gap directly motivates the relevance of a MMCP framework, which integrates a consistent decision matrix, hybrid data-driven weighting, complementary MCDM methods, consensus-based aggregation, and explicit sensitivity analysis, thereby ensuring a robust, reproducible, and methodologically sound global ranking.

Compared with previous MCDM-based MPPT evaluations, which typically rely on a single weighting scheme—either subjective or objective—combined with a single ranking method applied to two or three performance indicators, the proposed MMCP framework differs in three concrete respects. First, it combines two independent objective weighting schemes, CRITIC and entropy, through a data-driven fusion coefficient rather than a fixed or arbitrarily selected weight. Second, it cross-validates the resulting ranking using four complementary MCDM methods (TOPSIS, PROMETHEE II, VIKOR, and ELECTRE II) instead of a single method, and consolidates them through Borda–Copeland aggregation. Third, it explicitly quantifies ranking robustness through Spearman-based sensitivity analysis under $\pm 20\%$ perturbation of each criterion’s weight. To the authors’ knowledge, no prior comparative MPPT study integrates hybrid objective weighting, multi-method cross-validation, and formal robustness quantification into a single reproducible protocol, which is precisely the gap that MMCP addresses.

The main contributions of this work could be summarized as follows:

- **Unified comparative framework:** A consistent evaluation environment is established for multiple MPPT strategies (classical, intelligent, hybrid, and robust), using harmonized indicators such as efficiency, rise time, settling time, and error indices, including the integral absolute error (IAE), integral time absolute error (ITAE), and integral squared error (ISE).
- **Novel MMCP framework:** A complete decision-making architecture is proposed, from data preprocessing to consensus aggregation, going beyond traditional single-ranking approaches.
- **Hybrid objective weighting:** The combined use of entropy and CRITIC reduces subjectivity and enhances statistical reliability by accounting for both data dispersion and inter-criteria correlation.
- **Multi-method decision integration:** The simultaneous use of TOPSIS, PROMETHEE II, VIKOR, and ELECTRE II allows cross-validation of ranking results and detection of methodological inconsistencies.
- **Robust global ranking for decision support:** The final Borda–Copeland aggregation provides a stable and actionable ranking for selecting MPPT strategies under conflicting objectives (efficiency, speed, accuracy, stability, robustness).

2 Materials and Methods

2.1 MPPT Strategies and Dataset

The decision dataset, presented in Table 2, was constructed based on the consolidated comparative table provided in the Excel file, which compiles results reported across three journal articles and three conference papers dedicated to the evaluation of MPPT strategies within a homogeneous PV system architecture.

The set of alternatives covers four families of control strategies: conventional, metaheuristic-hybridized, intelligent, and robust approaches. The analyzed set includes the following MPPT strategies, covering different types of MPPT control approaches.

The six source studies were retained precisely because they share a common simulation environment: the same PV module (Jinko Tiger Neo 78HL4-(V), 620 W), the same boost-converter topology and electrical parameters (switching frequency $f = 5$ kHz, inductance $L = 0.3$ mH, capacitance $C = 342$ μ F, input capacitance $C_{in} = 500$ μ F, and load resistance $R = 60$ Ω) and the same six evaluation indicators (η_{MPPT} , t_r , t_s , IAE, ISE, ITAE) computed

under comparable irradiance/temperature test sequences. This methodological homogeneity, rather than a post-hoc harmonization of heterogeneous test conditions, is what justifies pooling their reported results into a single, formally consistent decision matrix.

Table 2. Comparative evaluation of Maximum Power Point Tracking (MPPT) algorithms

MPPT Method	Energy	Dynamic		Error		
	Energy Efficiency η_{MPPT} (%)	Rise Time t_r (ms)	Settling Time t_s (ms)	Integral Absolute Error (IAE)	Integral Squared Error (ISE)	Integral Time Absolute Error (ITAE)
P&O	97.37	44.60	64.0	252.5	2.252	85.12
P&O-WOA	98.44	31.46	48.0	238.3	2.016	84.55
P&O-SFLA	98.38	32.30	49.0	239.6	2.042	84.59
P&O-BAT	98.38	33.00	50.0	241.3	2.074	84.75
P&O-GWO	98.40	35.90	53.0	244.1	2.125	84.86
INC	97.37	57.40	75.0	264.9	2.390	85.00
INC-WOA	98.39	30.80	47.0	236.0	2.050	83.90
INC-GWO	98.35	37.40	52.0	245.0	2.050	84.30
INC-BAT	98.32	34.40	50.0	240.0	2.080	84.50
INC-SFLA	98.36	35.20	51.0	242.0	2.090	84.60
ANN pur	98.54	30.60	42.9	137.0	1.870	35.67
ANN-SMC	98.52	31.00	44.0	140.0	1.900	36.00
SMC1	96.95	44.55	64.0	103.2	1.542	13.58
SMC2	96.95	56.35	77.0	117.8	1.692	7.456
PSO	96.98	56.50	78.0	148.2	2.077	24.99
FL	99.04	29.90	41.0	130.0	1.516	31.20

Note: t_r = Rise Time; t_s = Settling Time; P&O-WOA = Perturb and Observe hybridized with Whale Optimization Algorithm; P&O-SFLA = Perturb and Observe hybridized with Shuffled Frog Leaping Algorithm; P&O-BAT = Perturb and Observe hybridized with Bat Algorithm; P&O-GWO = Perturb and Observe hybridized with Grey Wolf Optimizer; INC-WOA = Incremental Conductance hybridized with Whale Optimization Algorithm; INC-GWO = Incremental Conductance hybridized with Grey Wolf Optimizer; INC-BAT = Incremental Conductance hybridized with Bat Algorithm; INC-SFLA = Incremental Conductance hybridized with Shuffled Frog Leaping Algorithm; ANN-SMC = Artificial Neural Network enhanced Sliding Mode Control; SMC1 = sliding mode control 1st order; SMC2 = sliding mode control 2nd order; FL = Fuzzy Logic; PSO = Particle Swarm Optimization.

The comparative table is reformulated as a decision matrix $X = [x_{ij}]$, where i denotes the MPPT strategy and j the evaluation criterion. This reformulation transforms a set of comparative results into a formal MCDM problem, which is a necessary step prior to weighting, ranking, and robust aggregation.

2.2 Evaluated Methods

The alternatives are drawn from six complementary contributions. Hybrid P&O variants are extracted from a comparative study focused on bio-inspired metaheuristics; hybrid INC variants originate from a parallel study on Incremental Conductance hybridization; ANN, P&O-WOA, and INC-WOA approaches are taken from a comparative synthesis of conventional, hybrid, and intelligent strategies. Finally, ANN-SMC, sliding mode control 1st order (SMC1)/sliding mode control 2nd order (SMC2), and FL/PSO are derived from three conference papers addressing ANN-SMC hybridization, sliding mode controllers, and FL-PSO comparisons, respectively.

Six quantitative criteria, common to all selected methods, are considered:

- (1) η_{MPPT} (%), to be maximized;
- (2) t_r (ms), to be minimized;
- (3) t_s (ms), to be minimized;
- (4) IAE, to be minimized;
- (5) ISE, to be minimized;
- (6) ITAE, to be minimized.

These metrics jointly capture energy performance, transient dynamics, and tracking accuracy. In the source studies, t_r is defined over the 10–90% interval, t_s within a $\pm 5\%$ settling band, and the error indices, expressed by

Eq. (1), are computed based on the power tracking error:

$$e(t) = P_{\max}(t) - P_{pv}(t) \quad (1)$$

2.3 Multi-Method Convergence Protocol Process

This study introduces a novel multi-criteria selection method, termed the MMCP, designed to provide a robust and reproducible ranking of MPPT control strategies. The proposed method proceeds through five sequential stages:

I. Data acquisition and pre-processing: operational data are collected and prepared through cleaning, synchronization, outlier handling, and robust normalization, yielding a structured decision matrix.

II. Data-driven criteria weighting: objective criteria weights W are derived directly from the normalized matrix Z using a hybrid CRITIC–entropy weighting scheme, removing the need for subjective expert-based weighting.

III. Multi-method ranking (MCDM): the weighted matrix is processed through four complementary MCDM methods—TOPSIS, PROMETHEE II, VIKOR, and ELECTRE II—each generating an independent ranking of the alternatives.

IV. Ranking aggregation: the individual rankings are consolidated into a single global consensus ranking using Borda and Copeland aggregation rules, reducing the sensitivity of the final result to any single MCDM method.

V. Validation and robustness analysis: the consensus ranking is tested through concordance, consistency, and sensitivity analyses to confirm its stability across weighting schemes and ranking methods.

2.4 Data Acquisition and Pre-Processing

In accordance with Step 0 of the MMCP flowchart, the decision matrix was subjected to preprocessing to ensure both metric and statistical consistency. Four main operations were performed.

First, units were standardized: time values were expressed in milliseconds, efficiency in percentage, and error indices retained in their original units. A scaling inconsistency was corrected for PSO and FL, whose efficiency values were initially reported in decimal form and were converted to 96.98% and 99.04%, respectively.

Second, cross-source consistency was verified through a systematic comparison between the Excel dataset and the six original publications. This step ensured alignment of metrics, alternatives, and comparison assumptions. While perfect homogeneity across sources is not required, sufficient controlled compatibility is necessary to allow integration into a unified decision matrix.

Third, the data were subjected to robust normalization in order to make heterogeneous criteria, differing in scale and magnitude, comparable. Let z_{ij} denote the normalized value associated with alternative i and criterion j . For a benefit criterion, the adopted transformation is expressed by Eq. (2):

$$z_{ij} = \frac{x_{ij} - \min_i x_{ij}}{\max_i x_{ij} - \min_i x_{ij}} \quad (2)$$

whereas, for a cost criterion, the normalization is defined by Eq. (3):

$$z_{ij} = \frac{\max_i x_{ij} - x_{ij}}{\max_i x_{ij} - \min_i x_{ij}} \quad (3)$$

This formulation preserves the preferential orientation of the criteria while projecting all performances values onto a common interval $[0,1]$.

Fourth, an explicit orientation of the criteria was enforced: η_{MPPT} was treated as a benefit criterion, whereas t_r , t_s , IAE, ISE, and ITAE were treated as cost criteria. This distinction directly ensures the consistency of normalization, weight computation, and the preference mechanisms employed by MCDM methods.

2.5 Objective Criteria Weighting

In accordance with Step 1 of the flowchart, the criteria weights were not defined subjectively but derived from the data structure using a hybrid entropy–CRITIC scheme. The objective is to combine two complementary dimensions:

1. the informational dispersion of each criterion across alternatives;
2. its informational independence with respect to other criteria.

For the entropy component, the normalized matrix $Z = [z_{ij}]$ is first transformed into proportional values.

$$p_{ij} = \frac{z_{ij}}{\sum_{i=1}^m z_{ij}} \quad (4)$$

Then, the entropy of criterion j is defined by Eq. (5):

$$E_j = -k \sum_{i=1}^m p_{ij} \ln(p_{ij}) \quad (5)$$

where, $k = (\ln m)^{-1}$.

The corresponding degree of diversification is then expressed as: $d_j = 1 - E_j$.

For the CRITIC component, expressed by Eq. (6), the importance of criterion j is determined based on its standard deviation σ_j and its correlations with the other criteria:

$$C_j = \sigma_j \sum_{k=1}^n (1 - r_{jk}) \quad (6)$$

where r_{jk} denotes the correlation coefficient between criteria j and k .

The derived elementary weights are normalized in accordance with Eq. (7) and Eq. (8):

$$w_j^{(E)} = \frac{d_j}{\sum j d_j} \quad (7)$$

$$w_j^{(C)} = \frac{C_j}{\sum j C_j} \quad (8)$$

The final hybrid weight is obtained through a balanced additive fusion in accordance with Eq. (9):

$$w_j = \frac{1}{2} \left(w_j^{(E)} + w_j^{(C)} \right) \quad (9)$$

where, $\sum w_j = 1$.

This scheme makes it possible to reduce both the subjectivity of weighting and the redundancy effects between criteria, which are particularly important when several metrics describe partially correlated dynamic properties.

2.6 Multi-Method Ranking

In accordance with Step 2 of the flowchart, four complementary MCDM methods were employed to generate individual rankings and assess inter-method stability.

2.6.1 Technique for order preference by similarity to ideal solution

TOPSIS ranks the alternatives according to their closeness to the positive ideal solution and their distance from the negative ideal solution. After weighting, the Euclidean distance of each alternative i from the two ideal solutions is computed, followed by the closeness coefficient.

$$CC_i = \frac{D_i^-}{D_i^+ + D_i^-} \quad (10)$$

The closeness coefficient CC_i , calculated using Eq. (10), is used for ranking, with a higher value indicating a better alternative.

2.6.2 Preference ranking organization method for enrichment evaluation II

PROMETHEE II is based on a net outranking approach relying on pairwise preference comparisons across criteria. For each pair of alternatives, a preference function $P_j(a, b)$ is computed. The positive ϕ^+ , negative ϕ^- , and net $\phi = \phi^+ - \phi^-$ outranking flows are then used to establish a complete ranking.

2.6.3 VlseKriterijumska Optimizacija I Kompromisno Resenje

VIKOR seeks a compromise solution by combining the aggregated group utility and the maximum individual regret. The measures S_i and R_i are computed based on distances from the ideal solution and then combined into the index.

$$Q_i = v \frac{S_i - S^*}{S^- - S^*} + (1 - v) \frac{R_i - R^*}{R^- - R^*} \quad (11)$$

with $v = 0.5$ in this study, in order to assign equal importance to collective utility and maximum individual regret.

2.6.4 Elimination and choice translating reality II

ELECTRE II introduces outranking logic based on concordance and discordance indices. For each pair of alternatives (a, b) , an outranking relation is accepted if the weighted concordance level is sufficiently high and no significant discordance opposes this preference. This approach is particularly suitable when criteria are partially conflicting and not fully compensatory. The combined use of TOPSIS, PROMETHEE II, VIKOR, and ELECTRE II aims to avoid reliance on a single decision-making rationale when establishing the final ranking.

2.7 Ranking Aggregation by Simplified Multi-Method Convergence Protocol

In accordance with Step 3, individual rankings were aggregated to obtain a global consensus ranking.

2.7.1 Borda aggregation

The Borda rule assigns each alternative a score based on its position in each ranking. For M methods and N alternatives, the score of an alternative i is defined by Eq. (12):

$$B_i = \sum_{m=1}^M (N - r_i^{(m)}) \quad (12)$$

where, $r_i(m)$ denotes the rank of alternative i in method m . A higher B_i value indicates consistently favorable performance.

2.7.2 Copeland aggregation

The Copeland rule is based on pairwise comparisons. For each alternative, the score is computed as the difference between the number of wins and the number of losses across all pairwise comparisons induced by the individual rankings.

2.7.3 Global consensual ranking

The global ranking is derived from the comparison of Borda and Copeland scores. When both rules converge, the hierarchy is considered consensual; in cases of local divergence, the stability of top-ranked alternatives and rank consistency are examined in the subsequent step. This mechanism limits the influence of any single ranking method on the final decision.

2.8 Validation and Robustness Analysis

Step 4 aims to evaluate the robustness of the global ranking from an explicitly decision-oriented perspective. Four analyses were conducted:

First, inter-method concordance was assessed using Spearman and Kendall rank correlation coefficients to evaluate the level of agreement among TOPSIS, PROMETHEE II, VIKOR, and ELECTRE II.

Second, the consistency of the aggregated ranking was examined by comparing individual rankings with the consensus ranking.

Third, a sensitivity analysis to weights was performed by perturbing the weight vector W around its nominal values to assess the stability of dominant alternatives.

Fourth, a leave-one-criterion-out analysis was conducted to evaluate the impact of removing each criterion on the final ranking.

This step explicitly distinguishes the physical robustness of controllers—already addressed in the source studies—from the decision robustness of the ranking, which constitutes the specific contribution of the MMCP framework.

2.9 Performance Evaluation Protocol

The evaluation protocol is based on simulation results obtained under harmonized Matrix Laboratory (MATLAB)/Simulink environments, using a consistent PV–boost converter structure and standardized comparison conditions. The metrics in the decision matrix were extracted from three journal articles and three conference papers, selected only when they satisfied a common modeling baseline: the same PV generator type, identical boost converter configuration, consistent electrical parameters, and comparable performance indicators.

This homogeneity remains a controlled methodological assumption rather than a strict identity across all simulation campaigns. All MCDM processing and numerical analyses were implemented in Python using Jupyter Notebook, on a workstation equipped with an Intel Core i7 processor, 16 GB of RAM, and a 500 GB Solid-State Drive (SSD). This setup ensures full reproducibility of the weighting, ranking, aggregation, and validation processes.

Overall, the proposed methodology transforms a multi-source comparative dataset into a reproducible MCDM problem, in which MPPT alternatives are successively harmonized, objectively weighted, ranked using multiple decision logics, aggregated through consensus, and validated via robustness analysis. This architecture is designed to produce a traceable, stable, and methodologically sound global ranking, rather than conclusions dependent on a single metric or decision tool.

3 Results and Discussion

Section 3 presents the simulation results and their multi-criteria interpretation within the proposed MMCP framework. It is structured into two complementary levels: first, a descriptive analysis of the raw performance of MPPT strategies in terms of efficiency, dynamics, and accuracy; and second, a decision-oriented analysis based on hybrid weighting, individual MCDM rankings, consensus aggregation, and robustness testing.

This approach enables a transition from simple metric-based comparison to a global, stable, and methodologically justified hierarchy of the evaluated control strategies.

3.1 Normalized Decision Matrix

The normalized matrix Z represents the first critical transformation of the MPPT comparison problem, as it projects heterogeneous metrics, efficiency, dynamics, and integrated errors—into a common dimensionless space directly exploitable by MCDM methods.

At this stage, the interpretation does not yet correspond to a final ranking, but rather to a diagnostic perspective. It allows the identification of dominant performance profiles, structural trade-offs between alternatives, and zones of redundancy or conflict among criteria.

This step is consistent with the harmonized framework of the source studies, which systematically rely on η_{MPPT} , t_r , t_s , IAE, ISE, and ITAE as a common evaluation core.

Table 3 presents the normalized decision matrix Z as a color-coded heatmap, in which green cells denote relatively high normalized scores, red cells denote relatively low normalized scores, and yellow/orange cells correspond to intermediate values, with the gradient applied independently to each criterion column. This representation enables the relative strengths and weaknesses of each MPPT strategy to be readily identified prior to the weighting and aggregation stages. Examined under this representation, matrix Z reveals a clear structuring of the multi-criteria space of MPPT control strategies. The alternatives FL, standalone ANN, ANN-SMC, P&O–WOA, and INC–WOA occupy consistently favorable positions, characterized predominantly by green cells for η_{MPPT} , t_r , and t_s , reflecting a robust balance between energy efficiency, transient response speed, and tracking accuracy.

Table 3. Normalized indicator values of each MPPT method

Method	η_{MPPT}	t_r	t_s	IAE	ISE	ITAE
P&O	0.2009	0.4654	0.3783	0.0766	0.5789	0
P&O–WOA	0.7129	0.9432	0.8108	0.1645	0.4279	0.0073
P&O–SFLA	0.6842	0.9127	0.7837	0.1564	0.3981	0.0068
P&O–BAT	0.6842	0.8872	0.7567	0.1459	0.3615	0.0047
P&O–GWO	0.6937	0.7818	0.6756	0.1286	0.3032	0.0033
INC	0.2009	0	0.0810	0	0	0.0015
INC–WOA	0.6889	0.9672	0.8378	0.1787	0.3890	0.0157
INC–GWO	0.6698	0.7272	0.7027	0.1230	0.3890	0.0105
INC–BAT	0.6555	0.8363	0.7567	0.1539	0.3546	0.0079
INC–SFLA	0.6746	0.8072	0.7297	0.1416	0.3432	0.0066
ANN pur	0.7607	0.9745	0.9486	0.7909	0.5949	0.6367
ANN–SMC	0.7511	0.9600	0.9189	0.7724	0.5606	0.6324
SMC1	0	0.4672	0.3783	1.0000	0.9702	0.9211
SMC2	0	0.0381	0.0270	0.9097	0.7986	1.0000
PSO	0.0143	0.0327	0	0.7217	0.3581	0.7742
FL	1.0000	1.0000	1.0000	0.8342	1.0000	0.6942

Note: η_{MPPT} = MPPT Energy Efficiency; t_r = Rise Time; t_s = Settling Time; IAE = Integral Absolute Error; ISE = Integral Squared Error; ITAE = Integral Time Absolute Error; P&O–WOA = Perturb and Observe hybridized with Whale Optimization Algorithm; P&O–SFLA = Perturb and Observe hybridized with Shuffled Frog Leaping Algorithm; P&O–BAT = Perturb and Observe hybridized with Bat Algorithm; P&O–GWO = Perturb and Observe hybridized with Grey Wolf Optimizer; INC–WOA = Incremental Conductance hybridized with Whale Optimization Algorithm; INC–GWO = Incremental Conductance hybridized with Grey Wolf Optimizer; INC–BAT = Incremental Conductance hybridized with Bat Algorithm; INC–SFLA = Incremental Conductance hybridized with Shuffled Frog Leaping Algorithm; ANN–SMC = Artificial Neural Network enhanced Sliding Mode Control; SMC1 = sliding mode control 1st order; SMC2 = sliding mode control 2nd order; FL = Fuzzy Logic; PSO = Particle Swarm Optimization.

By contrast, P&O, INC, SMC1, SMC2, and PSO exhibit weaker or more polarized profiles: SMC1 and SMC2, for instance, combine red cells for η_{MPPT} with green cells for IAE, ISE, and ITAE, indicating pronounced trade-offs rather than uniformly poor performance. Within the hybrid families, WOA emerges as the most effective optimization operator for both P&O and INC, producing a marked shift toward green for the dynamic criteria t_r and

t_s relative to their baseline counterparts, whereas SFLA, BAT, and GWO yield more moderate, yellow-to-light-green improvements.

From an analytical standpoint, the heatmap highlights both a strong discriminative capacity between control families—evident in the contrast between predominantly green rows (FL and ANN-based controllers) and predominantly red or mixed-color rows (P&O, INC, SMC1, SMC2)—and a partial correlation among the dynamic and error-based criteria, particularly t_r , t_s , IAE, ISE, and ITAE, where similar color patterns recur jointly across several alternatives. This structure substantiates the adoption of a hybrid entropy–CRITIC objective weighting scheme, which is capable of capturing both informational dispersion and inter-criteria redundancy simultaneously. Accordingly, the normalization step constitutes not merely a rescaling operation but a structuring phase of the decision problem, the resulting color-coded representation providing an essential visual basis for the consistent, unbiased, and methodologically robust multi-criteria ranking developed in the subsequent sections.

3.2 Data-Driven Hybrid Objective Weighting

The CRITIC–entropy hybrid weighting, presented in Figure 1, confirms that the decision structure of the problem is primarily driven by the error-related criteria, which account for 53.65% of the total weight, followed by dynamic criteria (30.34%) and energy-related performance (16.02%). This result indicates that, within the considered normalized matrix, the discriminative power of the criteria is mainly carried by IAE, ISE, and especially ITAE, whereas η_{MPPT} contributes less significantly to the overall variability of the decision problem.

Having established the normalized decision matrix, we now turn to the weighting stage, whose purpose is to determine how much each criterion contributes to the overall ranking before any MCDM method is applied.

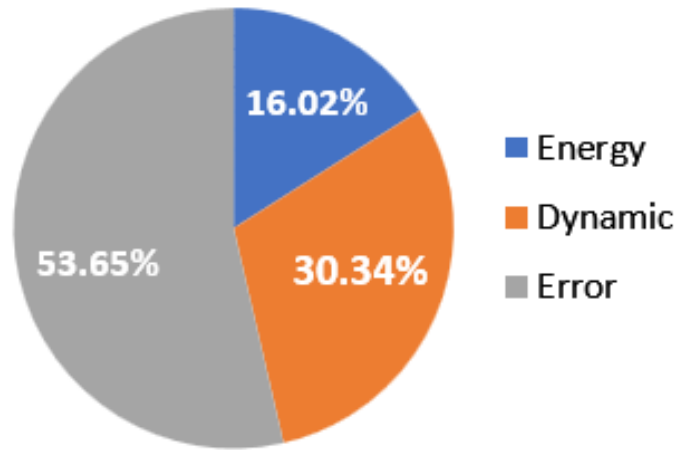


Figure 1. Weighted distribution of decision criteria

The hybrid weighting was computed by Eq. (13):

$$W_h = \alpha W_c + (1 - \alpha) W_e \quad (13)$$

with $\alpha = 0.9$, indicating a clear predominance of the CRITIC scheme in the final combination. As a result, the hybrid weights remain very close to the CRITIC weights, while still incorporating the informational contribution of entropy.

Unlike commonly adopted fixed or expert-assigned fusion coefficients, α is here computed directly from the data as $\alpha = \Sigma_j C_j / (\Sigma_j C_j + \Sigma_j d_j)$, i.e., the share of total discriminative information carried by the CRITIC scheme relative to the combined CRITIC–entropy information content. For the present decision matrix this yields $\alpha = 0.9014 (\approx 0.9)$, reflecting the fact that CRITIC—which jointly captures dispersion and inter-criteria conflict—extracts substantially more discriminative information from these six criteria than entropy alone, which only measures divergence from an equiprobable distribution. The subsequent sensitivity analysis (Section 3.6) confirms that the final ranking remains stable underweight perturbations well beyond the 10% margin attributable to the entropy component, indicating that the result is not an artifact of this specific α value.”

At the individual level, ITAE receives the highest weight ($W_h = 0.2483$), followed by IAE (0.1826), t_r (0.1604), η_{MPPT} (0.1602), and t_s (0.1430), and ISE (0.1056). This hierarchy shows that the weighting framework primarily emphasizes metrics reflecting cumulative error and overall temporal tracking quality, rather than instantaneous indicators such as speed or efficiency alone. In particular, the high weight assigned to ITAE highlights its strong discriminative role, as this criterion penalizes both the magnitude of the error and its persistence over

time. Conversely, ISE receives the lowest weight, suggesting a more limited discriminative capacity or partial redundancy with other error-related metrics.

From a methodological perspective, this weight distribution is consistent with an MPPT problem in which several alternatives exhibit relatively similar efficiency levels, while their transient behaviors and integrated errors remain more contrasted. The hybrid weighting, presented in Table 4, thus reduces the risk of artificially overemphasizing energy efficiency alone and refocuses the decision process on overall tracking quality, including stability, accuracy, and error persistence.

Table 4. Structure of hybrid weighting coefficients

Criterion	Family	CRITIC_Weight	Entropy_Weight	Hybrid_Weight
η_{MPPT} (%)	Energy	0.1644	0.1212	0.1601
t_r (ms)	Dynamic	0.1676	0.0943	0.1604
t_s (ms)	Dynamic	0.1484	0.0930	0.1429
IAE	Error	0.1820	0.1871	0.1825
ISE	Error	0.1084	0.0793	0.1056
ITAE	Error	0.2289	0.4248	0.2482

Note: η_{MPPT} = MPPT Energy Efficiency; t_r = Rise Time; t_s = Setting Time; IAE = Integral Absolute Error; ISE = Integral Squared Error; ITAE = Integral Time Absolute Error; CRITIC = Criteria Importance Through Intercriteria Correlation.

In summary, Step 1 indicates that the final ranking will be primarily driven by the control of cumulative dynamic errors, with transient dynamics occupying an intermediate position, while energy performance retains a significant but secondary role within the multi-criteria hierarchy.

With the hybrid weights fixed, the next step consists in translating these weights into an actual ordering of alternatives; TOPSIS is applied first as it offers the most direct geometric interpretation of closeness to an ideal solution

The Figure 2 shows that the hybrid weighting remains very close to the CRITIC scheme across all criteria, which is consistent with the high value of α . The contribution of entropy slightly adjusts the initial distribution without altering the overall hierarchy of weights.

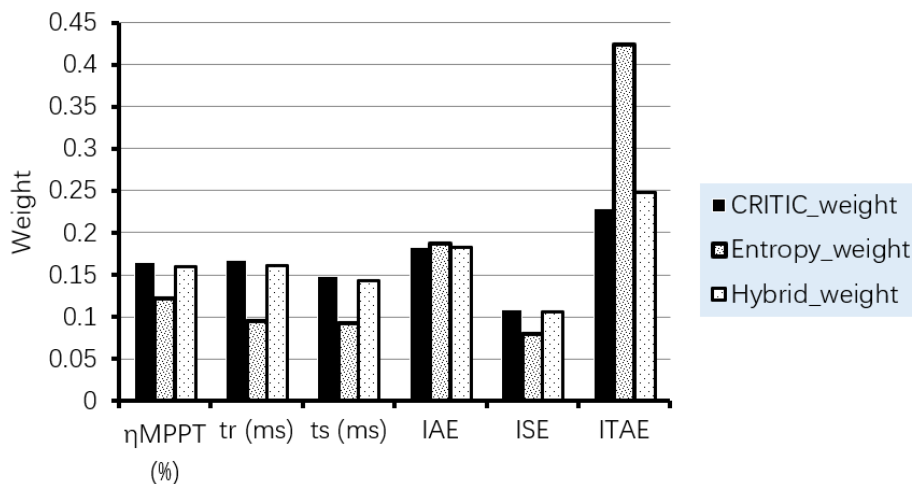


Figure 2. Comparative analysis of weighting coefficients across data-driven methods

Note: η_{MPPT} = MPPT Energy Efficiency; t_r = Rise Time; t_s = Setting Time; IAE = Integral Absolute Error; ISE = Integral Squared Error; ITAE = Integral Time Absolute Error; CRITIC = Criteria Importance Through Intercriteria Correlation.

ITAE remains the most influential criterion, exhibiting the largest deviation between entropy and CRITIC, while its hybrid weight continues to dominate. IAE also retains a high and stable weight across the three schemes. In contrast, ISE remains the least weighted criterion, reflecting a lower discriminative power or partial redundancy with other error indices. The criteria η_{MPPT} , t_r , and t_s occupy intermediate positions, with a slight revaluation of dynamic performance relative to energy efficiency.

In summary, the hybrid scheme preserves the structural robustness of CRITIC while incorporating the informational contribution of entropy. This results in a more balanced weighting, still dominated by cumulative error criteria—particularly ITAE—which is expected to strongly drive the final ranking.

3.3 Results and Analysis of the Multi-Criteria Decision-Making

3.3.1 Technique for order preference by similarity to ideal solution

The TOPSIS profile, shows in Figure 3, confirms a hierarchy consistent with the weighted decision matrix. Closeness to the ideal solution is highest for FL, followed by ANN–SMC and standalone ANN, reflecting both a minimal distance to the positive ideal solution (D^+) and a significant distance from the negative ideal solution (D^-). Conversely, P&O, INC, and, to a lesser extent, PSO exhibit the lowest closeness coefficients, confirming their limited competitiveness in a multi-criteria context.

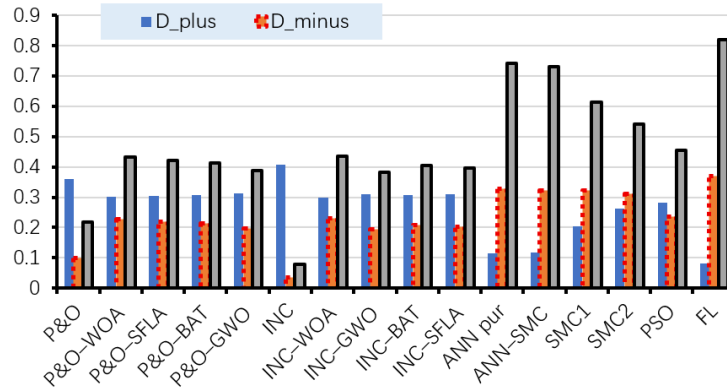


Figure 3. Evolution of relative distances and closeness coefficient according to the TOPSIS method

Note: TOPSIS = Technique for Order Preference by Similarity to Ideal Solution; D_{plus} = Euclidean distance from positive ideal solution; D_{minus} = Euclidean distance from negative ideal solution; P&O–WOA = Perturb and Observe hybridized with Whale Optimization Algorithm; P&O–SFLA = Perturb and Observe hybridized with Shuffled Frog Leaping Algorithm; P&O–BAT = Perturb and Observe hybridized with Bat Algorithm; P&O–GWO = Perturb and Observe hybridized with Grey Wolf Optimizer; INC–WOA = Incremental Conductance hybridized with Whale Optimization Algorithm; INC–GWO = Incremental Conductance hybridized with Grey Wolf Optimizer; INC–BAT = Incremental Conductance hybridized with Bat Algorithm; INC–SFLA = Incremental Conductance hybridized with Shuffled Frog Leaping Algorithm; ANN–SMC = Artificial Neural Network enhanced Sliding Mode Control; SMC1 = sliding mode control 1st order; SMC2 = sliding mode control 2nd order; FL = Fuzzy Logic; PSO = Particle Swarm Optimization.

The evolution of the curves also reveals a clear break between conventional methods and intelligent or hybrid approaches. The alternatives based on standalone ANN, ANN–SMC, and FL exhibit a sharp increase in closeness, mainly driven by a significant reduction in D^+ , while WOA-based hybrid variants occupy a favorable intermediate position. This configuration indicates that the top-ranked solutions are not only strong on individual criteria but also achieve a more balanced overall proximity to the multi-criteria ideal.

In summary, the Figure 3 highlights a robust dominance of FL, followed by the ANN–SMC/standalone ANN group, then the WOA-based hybrids, while P&O and INC remain structurally dominated. The behavior of the closeness coefficient thus confirms the ability of TOPSIS to clearly discriminate alternatives based on their global trade-off between efficiency, dynamics, and error performance.

3.3.2 Preference ranking organization method for enrichment evaluation II

The PROMETHEE II profile confirms a globally consistent hierarchy, while revealing greater sensitivity in the intermediate range compared to TOPSIS. The highest net flows ϕ are observed for FL, standalone ANN, and ANN–SMC, confirming their multi-criteria superiority. In contrast, INC and P&O are associated with strongly negative net flows, reflecting their structural domination by most of the other alternatives.

The joint interpretation of ϕ^+ , ϕ^- , and ϕ , presented in Figure 4, shows that the best strategies combine a high positive flow with a low negative flow, reflecting a strong outranking capability. This is particularly evident for FL and the ANN/ANN–SMC group. The INC–WOA and P&O–WOA variants maintain a favorable but less dominant profile, confirming their position as strong intermediate alternatives. In contrast, PSO, SMC1, and SMC2 exhibit more contrasted flows, indicating a higher sensitivity to the specific preference structure of PROMETHEE.

In summary, the Figure 4 highlights a robust dominance of intelligent methods, followed by WOA-based hybrid approaches, while conventional controllers remain clearly dominated. The more pronounced dispersion of net flows in the middle of the ranking confirms that PROMETHEE II provides a finer discrimination of intermediate solutions, making it particularly suitable for assessing local ranking stability prior to consensus aggregation.

3.3.3 Multi-criteria optimization and compromise solution

The VIKOR results, presented in Figure 5, reveal a clear hierarchy among the alternatives. FL exhibits the lowest values of S, R, and Q, confirming its position as the best compromise solution. Standalone ANN and

ANN–SMC follow closely, with similarly low values, indicating a strong balance between overall performance and maximum regret. In contrast, P&O and especially INC show the highest values, confirming their weak multi-criteria competitiveness.

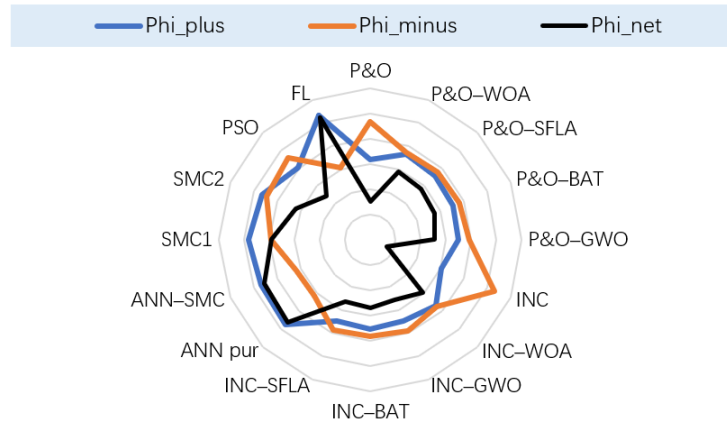


Figure 4. Comparative analysis of Φ^+ , Φ^- , and Φ flows using the PROMETHEE II method

Note: PROMETHEE II = Preference Ranking Organization Method for Enrichment Evaluation II; Phi_plus = Positive outranking flows; Phi_minus = Negative outranking flows; Phi_net = Net outranking flows; P&O–WOA = Perturb and Observe hybridized with Whale Optimization Algorithm; P&O–SFLA = Perturb and Observe hybridized with Shuffled Frog Leaping Algorithm; P&O–BAT = Perturb and Observe hybridized with Bat Algorithm; P&O–GWO = Perturb and Observe hybridized with Grey Wolf Optimizer; INC–WOA = Incremental Conductance hybridized with Whale Optimization Algorithm; INC–GWO = Incremental Conductance hybridized with Grey Wolf Optimizer; INC–BAT = Incremental Conductance hybridized with Bat Algorithm; INC–SFLA = Incremental Conductance hybridized with Shuffled Frog Leaping Algorithm; ANN–SMC = Artificial Neural Network enhanced Sliding Mode Control; SMC1 = sliding mode control 1st order; SMC2 = sliding mode control 2nd order; FL = Fuzzy Logic; PSO = Particle Swarm Optimization.

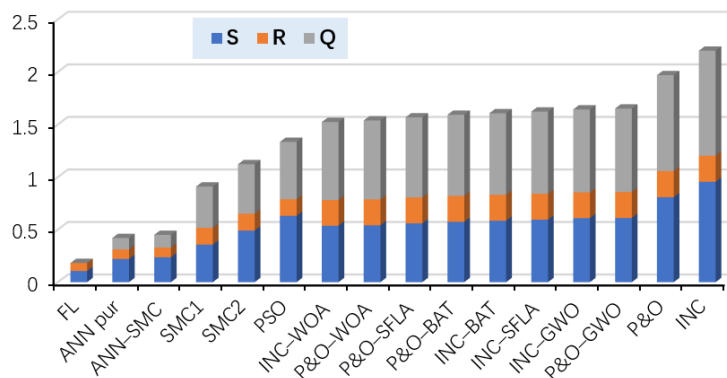


Figure 5. Comparative evaluation of alternatives using the S, R, and Q indices under the VIKOR method

Note: S = Values of group utility (VIKOR); R = Values of individual regret (VIKOR); Q = VIKOR compromise index; VIKOR = Multi-Criteria Optimization and Compromise Solution; P&O–WOA = Perturb and Observe hybridized with Whale Optimization Algorithm; P&O–SFLA = Perturb and Observe hybridized with Shuffled Frog Leaping Algorithm; P&O–BAT = Perturb and Observe hybridized with Bat Algorithm; P&O–GWO = Perturb and Observe hybridized with Grey Wolf Optimizer; INC–WOA = Incremental Conductance hybridized with Whale Optimization Algorithm; INC–GWO = Incremental Conductance hybridized with Grey Wolf Optimizer; INC–BAT = Incremental Conductance hybridized with Bat Algorithm; INC–SFLA = Incremental Conductance hybridized with Shuffled Frog Leaping Algorithm; ANN–SMC = Artificial Neural Network enhanced Sliding Mode Control; SMC1 = sliding mode control 1st order; SMC2 = sliding mode control 2nd order; FL = Fuzzy Logic; PSO = Particle Swarm Optimization.

The intermediate range displays a coherent structure: WOA-based hybrid approaches occupy the best positions after the leading group, while BAT, SFLA, and GWO remain less competitive but still outperform conventional methods. PSO, SMC1, and SMC2 appear in intermediate positions with relatively lower robustness, reflecting acceptable but non-dominant trade-offs.

In summary, the Q profile confirms that FL, followed by standalone ANN and ANN–SMC, represent the most favorable solutions according to the VIKOR compromise logic, while P&O and INC remain structurally dominated. The Figure 5 thus validates a hierarchy that is globally consistent with TOPSIS and PROMETHEE, while reinforcing

the robustness of the leading group.

3.3.4 Elimination and choice translating reality II

The ELECTRE II profile reveals a more contrasted hierarchy compared to the other methods, based on a net outranking logic. FL exhibits the highest net concordance score, followed by standalone ANN and ANN–SMC, confirming their ability to dominate a large number of alternatives across most criteria. In contrast, INC and P&O display the most negative values, reflecting structural domination and weak multi-criteria competitiveness.

The intermediate range, presented in Figure 6, confirms the strong performance of WOA-based hybrid approaches, particularly INC–WOA and P&O–WOA, which maintain positive net concordance scores and are positioned just behind the leading group. In contrast, PSO, INC–GWO, INC–SFLA, and P&O–GWO exhibit negative net concordance values, indicating a weaker outranking capability despite sometimes acceptable performance in other methods.

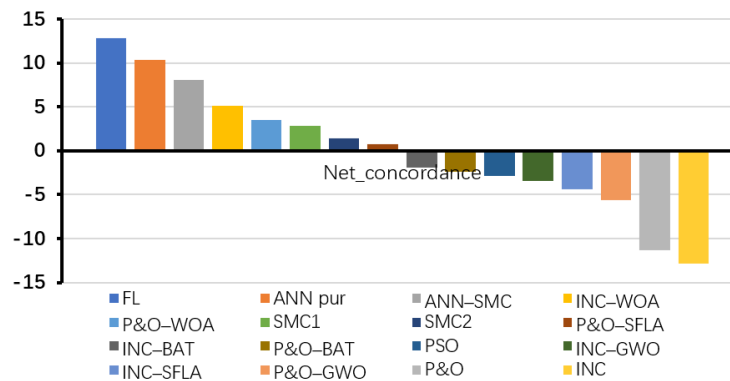


Figure 6. Distribution of net concordance among alternatives using the ELECTRE II method

Note: ELECTRE II = Elimination and Choice Translating Reality II; P&O–WOA = Perturb and Observe hybridized with Whale Optimization Algorithm; P&O–SFLA = Perturb and Observe hybridized with Shuffled Frog Leaping Algorithm; P&O–BAT = Perturb and Observe hybridized with Bat Algorithm; P&O–GWO = Perturb and Observe hybridized with Grey Wolf Optimizer; INC–WOA = Incremental Conductance hybridized with Whale Optimization Algorithm; INC–GWO = Incremental Conductance hybridized with Grey Wolf Optimizer; INC–BAT = Incremental Conductance hybridized with Bat Algorithm; INC–SFLA = Incremental Conductance hybridized with Shuffled Frog Leaping Algorithm; ANN–SMC = Artificial Neural Network enhanced Sliding Mode Control; SMC1 = sliding mode control 1st order; SMC2 = sliding mode control 2nd order; FL = Fuzzy Logic; PSO = Particle Swarm Optimization.

In summary, ELECTRE II confirms the robust dominance of FL, followed by the standalone ANN/ANN–SMC group, while reinforcing the relative superiority of WOA-based hybrid approaches. The pronounced dispersion of net scores highlights the strong discriminative power of this method, clearly distinguishing dominant, intermediate, and structurally dominated solutions.

3.3.5 Aggregation

The aggregation results show strong consistency between Borda, Copeland, and the final score. FL retains the top position with the highest final score, followed by standalone ANN and ANN–SMC, confirming the stability of the leading group already observed in the individual rankings. At the opposite end, P&O and especially INC remain in the last positions, reflecting structural domination independent of the aggregation rule.

Because the four MCDM methods can in principle disagree on marginal cases, the following step aggregates their individual rankings into a single consensus order, which is the ranking ultimately recommended for decision support

The intermediate range, presented in Figure 7, shows that aggregation primarily plays a role in stabilizing local rankings. WOA-based hybrid approaches (INC–WOA and P&O–WOA) maintain high scores and remain positioned just behind intelligent methods, while SMC1, SMC2, and PSO occupy intermediate positions that are more sensitive to the ranking method. The differences between Borda and Copeland scores remain generally limited for the top alternatives, confirming a strong robustness of the consensus.

In summary, the consensus-based aggregation does not alter the overall ranking structure but enhances its clarity and robustness by consolidating dominant positions and reducing instability in the intermediate range. The final score thus validates a clear hierarchy: FL dominates, followed by the ANN/ANN–SMC group, WOA-based hybrid approaches form the strongest intermediate cluster, while P&O and INC remain structurally dominated.

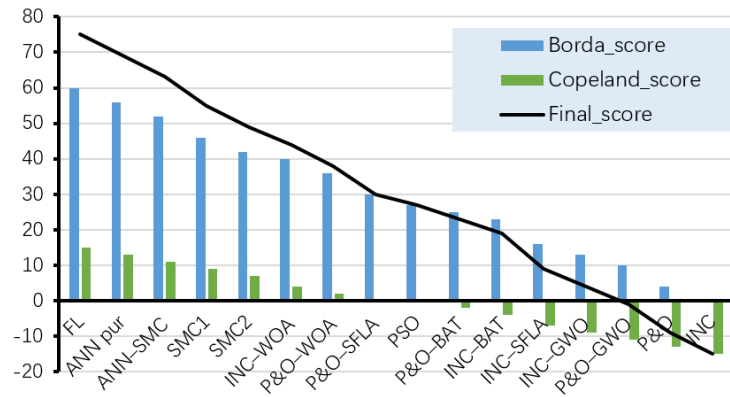


Figure 7. Comparative analysis of aggregated ranking scores

Note: P&O-WOA = Perturb and Observe hybridized with Whale Optimization Algorithm; P&O-SFLA = Perturb and Observe hybridized with Shuffled Frog Leaping Algorithm; P&O-BAT = Perturb and Observe hybridized with Bat Algorithm; P&O-GWO = Perturb and Observe hybridized with Grey Wolf Optimizer; INC-WOA = Incremental Conductance hybridized with Whale Optimization Algorithm; INC-GWO = Incremental Conductance hybridized with Grey Wolf Optimizer; INC-BAT = Incremental Conductance hybridized with Bat Algorithm; INC-SFLA = Incremental Conductance hybridized with Shuffled Frog Leaping Algorithm; ANN-SMC = Artificial Neural Network enhanced Sliding Mode Control; SMC1 = sliding mode control 1st order; SMC2 = sliding mode control 2nd order; FL = Fuzzy Logic; PSO = Particle Swarm Optimization.

3.4 Validation

The concordance radar indicates a high level of agreement among the MCDM methods. Spearman and Kendall coefficients generally range between approximately 0.88 and 0.97, indicating that the rankings produced by TOPSIS, PROMETHEE, VIKOR, and ELECTRE are globally consistent. The strongest correlation is observed between TOPSIS and VIKOR, suggesting a close alignment between the distance-to-ideal and compromise-based decision logics in this problem. In contrast, method pairs involving ELECTRE—and to a lesser extent PROMETHEE—show slightly larger deviations, reflecting the sensitivity of outranking approaches to certain intermediate alternatives.

From a methodological perspective, these results, shown in Figure 8, confirm that inter-method uncertainty remains limited and is primarily concentrated in the intermediate ranks, without challenging the overall structure of the ranking. The high level of concordance thus justifies the final consensus-based aggregation: rather than combining conflicting hierarchies, it consolidates a core set of already strongly convergent results. In this sense, the Figure 8 validates the decision robustness of the MMCP framework and reinforces the credibility of the resulting global ranking.

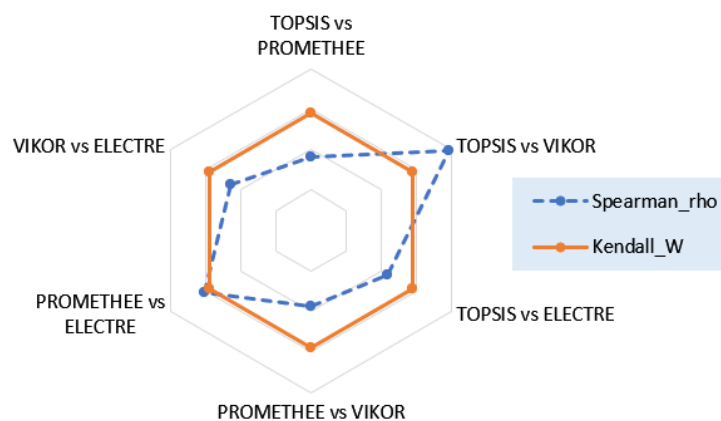


Figure 8. Inter-method comparison of Spearman and Kendall coefficients

Note: TOPSIS = Technique for Order Preference by Similarity to Ideal Solution; VIKOR = VlseKriterijumska Optimizacija I Kompromisno Resenje; ELECTRE = Elimination and Choice Translating Reality; PROMETHEE = Preference Ranking Organization Method for Enrichment Evaluation.

The Figure 9 confirms a strong convergence between the final ranking and the individual rankings. Points located on or near the diagonal indicate that, for most alternatives, the consensus ranking closely reproduces the common trend identified by TOPSIS, VIKOR, PROMETHEE, and ELECTRE. This consistency is particularly evident at the

extremes: FL, standalone ANN, and ANN–SMC consistently rank at the top, while P&O and INC remain at the bottom. The most noticeable deviations occur in the intermediate range, mainly for PSO, SMC1/SMC2, and certain metaheuristic hybrid approaches. These differences reflect local sensitivity to the compromise and outranking logics specific to each method, without affecting the overall structure of the hierarchy.

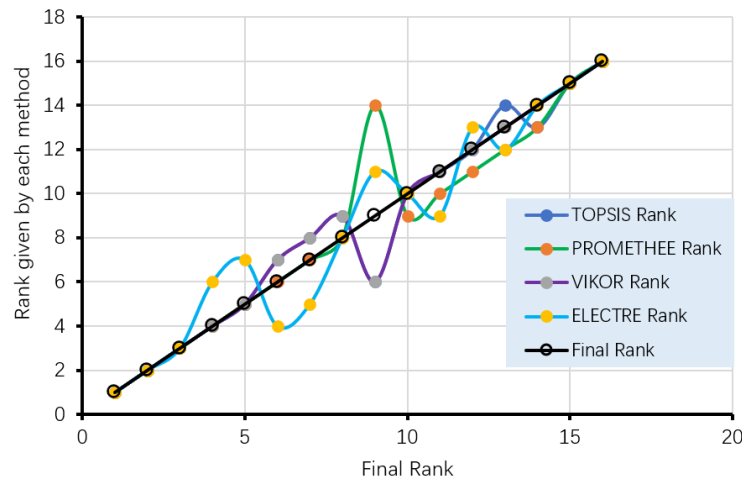


Figure 9. Comparative study of rankings derived from MCDM methods and the final ranking.

Note: MCDM = Multi-Criteria Decision-Making; TOPSIS = Technique for Order Preference by Similarity to Ideal Solution; VIKOR = VlseKriterijumska Optimizacija I Kompromisno Resenje; ELECTRE = Elimination and Choice Translating Reality; PROMETHEE = Preference Ranking Organization Method for Enrichment Evaluation.

The final ranking acts as a stabilized consensus: it preserves the robust trends shared across the four methods while absorbing local fluctuations observed in intermediate positions. This confirms the reliability of the MMCP aggregation and the methodological robustness of the global ranking.

3.5 Methodological Implications

The MMCP framework offers methodological value that extends beyond the specific context of PV MPPT. By combining data-driven objective weighting, multiple ranking methods, consensus-based aggregation, and robustness validation, it provides a generic framework for multi-criteria selection problems involving trade-offs between performance, speed, stability, and accuracy.

This architecture can be readily extended to other advanced control applications, including power converters, microgrids, electromechanical systems, and industrial processes, where multiple competing strategies must be evaluated based on heterogeneous and potentially conflicting criteria. From this perspective, the simplified MMCP framework not only delivers a robust ranking of MPPT strategies but also establishes a reproducible and extensible decision-making procedure for comparative evaluation in complex environments.

4 Conclusion

This study proposed a MMCP framework for the robust ranking of MPPT control strategies in PV systems within a unified decision-making context. Unlike conventional mono-criterion or single-method comparisons, the proposed framework integrates robust data preprocessing, hybrid objective weighting (CRITIC–entropy), multiple complementary MCDM methods (TOPSIS, PROMETHEE II, VIKOR, and ELECTRE II), and consensus-based rank aggregation (Borda–Copeland), thereby reducing methodological bias and enhancing ranking stability. The results demonstrate that the decision structure is primarily driven by dynamic error-related criteria, with the error family accounting for the largest share of the hybrid weight, followed by dynamics and energy efficiency. Under this configuration, FL, ANN, and ANN–SMC consistently form the leading group across all MCDM methods and in the final consensus ranking, while WOA-based hybrid approaches emerge as the most competitive intermediate alternatives. In contrast, conventional P&O and INC controllers remain structurally dominated. The high level of agreement among the ranking methods confirms that the final ordering is not an artifact of a single decision logic but rather the outcome of a stable and convergent multi-criteria evaluation process.

From an engineering perspective, the MMCP framework provides a traceable and reproducible tool for MPPT selection in the presence of multiple conflicting objectives, including efficiency, transient response, and tracking accuracy. It therefore offers practical value for both research benchmarking and industrial decision support in real PV systems. Moreover, the framework can be extended to other control-selection problems involving heterogeneous and partially conflicting performance indicators.

This study is nonetheless subject to several limitations. First, the decision matrix relies on performance indicators reported across six independent published studies rather than a single, unified experimental campaign; although the underlying PV–converter configuration and evaluation indicators are consistent across sources (Section 2.1), residual differences in test conditions across the original studies cannot be entirely excluded. Second, the analysis is conducted entirely in simulation, and no real-time hardware-in-the-loop or experimental prototype validation has yet been performed; controller behavior under sensor noise, communication latency, or component non-idealities may therefore differ from the reported values. Third, the CRITIC and entropy weighting schemes assume that criteria contribute independently to the hybrid weight, whereas the correlation analysis (Section 3.1) shows that some criteria, notably t_r , t_s , IAE, ISE, and ITAE, are partially correlated; while the sensitivity analysis indicates that the final ranking is robust to this simplification, a formal treatment of criteria correlation in the weighting stage is left for future work. Addressing these points, in particular through hardware-in-the-loop validation of the top-ranked strategies, constitutes a natural extension of the present framework

Future work should extend this framework by incorporating implementation complexity, computational cost, and qualitative criteria, as well as by conducting real-time experimental validation under practical operating conditions. These extensions would further enhance the applicability of MMCP as a robust decision-support methodology for advanced energy control systems.

Author Contributions

Conceptualization, W.O.A. and M.G.; methodology, W.O.A.; software, W.O.A.; validation, W.O.A., N.B., and M.G.; formal analysis, W.O.A.; investigation, W.O.A.; resources, W.O.A.; data curation, W.O.A.; writing—original draft preparation, W.O.A.; writing—review and editing, N.B. and M.G.; visualization, W.O.A.; supervision, M.G.; project administration, M.G. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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Nomenclature

Maximum Power Point Tracking (MPPT) Control Strategies

Abbreviation	Full Designation	Notes/Context
P&O	Perturb and Observe	Conventional MPPT method
INC	Incremental Conductance	Conventional MPPT method
WOA	Whale Optimization Algorithm	Metaheuristic optimizer
SFLA	Shuffled Frog Leaping Algorithm	Metaheuristic optimizer
BAT	Bat Algorithm	Metaheuristic optimizer
GWO	Grey Wolf Optimizer	Metaheuristic optimizer
PSO	Particle Swarm Optimization	Metaheuristic optimizer
ANN	Artificial Neural Network	Intelligent MPPT controller
FL	Fuzzy Logic	Intelligent MPPT controller
SMC	Sliding Mode Control	Robust control strategy
SMC1	1st-Order Sliding Mode Control	Robust MPPT controller
SMC2	2nd-Order Sliding Mode Control	Robust MPPT controller
ANN–SMC	Artificial Neural Network enhanced Sliding Mode Control	Hybrid intelligent/robust strategy
P&O–WOA	Perturb and Observe hybridized with Whale Optimization Algorithm	Hybrid metaheuristic strategy
P&O–SFLA	Perturb and Observe hybridized with Shuffled Frog Leaping Algorithm	Hybrid metaheuristic strategy
P&O–BAT	Perturb and Observe hybridized with Bat Algorithm	Hybrid metaheuristic strategy
P&O–GWO	Perturb and Observe hybridized with Grey Wolf Optimizer	Hybrid metaheuristic strategy
INC–WOA	Incremental Conductance hybridized with Whale Optimization Algorithm	Hybrid metaheuristic strategy
INC–GWO	Incremental Conductance hybridized with Grey Wolf Optimizer	Hybrid metaheuristic strategy
INC–BAT	Incremental Conductance hybridized with Bat Algorithm	Hybrid metaheuristic strategy

INC–SFLA	Incremental Conductance hybridized with Shuffled Frog Leaping Algorithm	Hybrid metaheuristic strategy
Multi-Criteria Decision-Making (MCDM) Methods		
Abbreviation	Full Designation	Notes/Context
MCDM	Multi-Criteria Decision-Making	General methodological family
MMCP	Multi-Method Convergence Protocol	Proposed framework in this study
TOPSIS	Technique for Order Preference by Similarity to Ideal Solution	Distance-based MCDM method
PROMETHEE II	Preference Ranking Organization Method for Enrichment Evaluation II	Outranking MCDM method
VIKOR	Vlsekriterijumska Optimizacija I Kompromisno Resenje	Compromise-based MCDM method
ELECTRE II	ELimination Et Choix Traduisant la REalite II	Concordance/discordance MCDM method
CRITIC	CRiteria Importance Through Intercriteria Correlation	Objective weighting method
Performance Metrics and Criteria		
Abbreviation	Full Designation	Notes/Context
MPPT	Maximum Power Point Tracking	Core control objective
η_{MPPT}	MPPT Energy Efficiency	Benefit criterion, expressed in %
t_r	Rise Time	Cost criterion, expressed in ms
t_s	Settling Time	Cost criterion, expressed in ms
IAE	Integral Absolute Error	Cost criterion
ISE	Integral Squared Error	Cost criterion
ITAE	Integral Time Absolute Error	Cost criterion - highest hybrid weight
$e(t)$	Power tracking error	$e(t) = P_{\text{max}}(t) - P_{\text{pv}}(t)$
Mathematical Variables and Notation		
Abbreviation	Full Designation	Notes/Context
$X = [x_{ij}]$	Raw decision matrix	i : alternative, j : criterion
$Z = [z_{ij}]$	Normalized decision matrix	Values in $[0, 1]$
p_{ij}	Proportional value of z_{ij} for entropy computation	Used in Shannon entropy formula
E_j	Shannon entropy of criterion j	Informational dispersion measure
k	Entropy normalization constant	$k = (\ln m)^{-1}$
d_j	Degree of diversification of criterion j	$d_j = 1 - E_j$
σ_j	Standard deviation of criterion j	Used in CRITIC formula
r_{jk}	Pearson correlation coefficient between criteria j and k	Used in CRITIC formula
C_j	CRITIC importance coefficient of criterion j	
$w_j^{(E)}$	Normalized Entropy weight of criterion j	
$w_j^{(C)}$	Normalized CRITIC weight of criterion j	
w_j/W_h	Final hybrid weight of criterion j	$\sum w_j = 1$
α	Fusion coefficient in hybrid weighting	$\alpha = 0.9$ in this study
W_c	CRITIC weight vector	Component of hybrid weight
W_e	Entropy weight vector	Component of hybrid weight
CC_i	Closeness Coefficient (TOPSIS)	Higher value $\rightarrow$ better alternative
D_i^+	Euclidean distance from positive ideal solution	TOPSIS metric
D_i^-	Euclidean distance from negative ideal solution	TOPSIS metric
$P_j(a, b)$	Pairwise preference function (PROMETHEE II)	
ϕ^+, ϕ^-, ϕ	Positive, negative, and net outranking flows	PROMETHEE II metrics
S_i, R_i	Group utility and maximum individual regret (VIKOR)	
Q_i	VIKOR compromise index	$v = 0.5$ in this study
v	Weight of strategy of maximum group utility (VIKOR)	$v = 0.5$ in this study

S^*, S^-	Best and worst values of group utility (VIKOR)	
R^*, R^-	Best and worst values of individual regret (VIKOR)	
B_i	Borda score of alternative i	Higher value $\rightarrow$ more favorable
$r_i(m)$	Rank of alternative i in method m	Used in Borda formula
M	Number of MCDM methods	$M = 4$ in this study
N	Number of alternatives	$N = 16$ in this study
m	Number of alternatives (entropy context)	Used in entropy formula
n	Number of criteria	$n = 6$ in this study

System Parameters and Hardware

Abbreviation	Full Designation	Notes/Context
PV	Photovoltaic	Solar energy conversion system
PSC	Partial Shading Condition	Non-uniform irradiance scenario
GMPP	Global Maximum Power Point	Under partial shading conditions
DC-DC	Direct Current to Direct Current converter	Boost converter topology used
f	Switching frequency	$f = 5$ kHz
L	Inductance of boost converter	$L = 0.3$ mH
C	Output capacitance of boost converter	$C = 342$ μ F
C_{in}	Input capacitance of boost converter	$C_{in} = 500$ μ F
R	Load resistance	$R = 60$ Ω
$P_{max}(t)$	Maximum available PV power at time t	
$P_{pv}(t)$	Actual PV output power at time t	

Statistical Validation Indices

Abbreviation	Full Designation	Notes/Context
W (Kendall)	Kendall's coefficient of concordance	$W = 0.9449$ in this study
ρ_s	Spearman rank correlation coefficient	Range: 0.891–0.997 in this study
τ (Kendall τ)	Kendall's rank correlation coefficient	Inter-method agreement metric

Abbreviation	Full Designation	Notes/Context
MATLAB	MATrix LABoratory	Simulation environment used
Simulink	Simulation and Model-Based Design tool	Graphical MATLAB extension
SSD	Solid-State Drive	Storage hardware
RAM	Random Access Memory	Computing resource